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NovaConnect: An AI-Integrated Academic Collaboration and Resource Management Platform

THEME: INNOVATION BEYOND BOUNDARIES (EDUCATION ECOSYSTEM)

ABSTRACT:

With the increasing digitalization of educational institutions, academic portals are widely used for managing attendance, results, and administrative data. However, many institutional portals lack structured communication systems and intelligent academic resource coordination. Informal messaging platforms are often used for student–faculty interaction, leading to unorganized discussions, data loss, and reduced academic accountability. To address these challenges, this project proposes NovaConnect, an AI-integrated academic collaboration and resource management platform designed as an extension to the institutional portal.

The primary objective of the system is to create a secure, domain-authenticated environment that enables structured communication, centralized resource sharing, assignment management, and attendance tracking within a single ecosystem. The platform incorporates group and private messaging, subject-wise discussion forums, faculty-controlled note uploads, and digital assignment submission.

The proposed solution follows a three-tier architecture consisting of a frontend interface, backend application server, and relational database system. Artificial Intelligence techniques are integrated to enhance functionality through automatic forum query categorization, AI-based note summarization, assignment similarity detection, and student engagement analytics.

The innovative aspect of NovaConnect lies in combining secure institutional authentication with intelligent academic assistance and structured communication, rather than functioning as a standalone messaging system. By integrating communication, academic resources, and data-driven insights, the platform improves transparency, operational efficiency, and academic coordination.

KEYWORDS:

Academic Collaboration, Artificial Intelligence, Domain Authentication, Educational Portal, Resource Management, Student Engagement, Real-Time Communication.